

Section : Biology

Q.1 Recombinant DNA is directly injected into the nucleus of an animal cell.
This method is known as-

1. Biolistics
2. Transformation
3. Micro-injection
4. Insertion

Options 1. 1

2. 2
3. 3
4. 4

Q.2 What does 'r' in equation $dN/dt = rN (K-N / K)$ represent ?

1. Extensive rate of natural increase
2. Intrinsic rate of natural increase
3. Extensive rate of natural decrease
4. Intrinsic rate of natural decrease

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Which of the following steps are included to make a healthy society?

- (A) Aware the people to uncontrolled population growth
(B) Aware the people to social evil like sex abuse
(C) Aware the people to sex related crime
(D) The Peoples lead to increase the mortality rate of mother and infant
Choose the correct answer from the options given below:

1. (A),(C) and (D) only
2. (A),(B)and(C)only
3. (A)and (D)only
4. (A),(B)and(D)only

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Failure to segregation of chromatids during cell division cycle which involves gain or loss of chromosome(s) is called:

1. Polyploidy
2. Euploidy
3. Aneuploidy
4. Monosomy

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Which one of the following(s) is/ are inverted pyramid in an ecosystem?

- (A) Pyramid of energy
(B) Pyramid of biomass (land)
(C) Pyramid of biomass (sea)
(D) Pyramid of Number
Choose the correct answer :

1. (A), (B) and (D) only
2. (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3

Q.6 The specific DNA sequence which is responsible for initiation of replication is known as :

1. Plasmid
2. Recombinant DNA
3. Template DNA
4. Origin of replication (ori)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Match List-I with List-II

List-I

List-II

Enzyme

Function

(A) Nuclease

(I) Synthesizes DNA from RNA

(B) Ligase

(II) Have both directions exonuclease activity and synthesize new polynucleotide

(C) DNA Polymerase I

(III) Degrades nucleic acids by breaking phosphodiester bonds

(D) Reverse Transcriptase

(IV) Joins DNA molecules together by formation of phosphodiester bonds

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Biopiracy refers to :

1. Use of bioresources by companies with proper authorization
2. Copying of someone's work without consent
3. Use of bio-resources by companies and other organizations without proper authorization from countries and people concerned without compensation amount
4. Use of bio-resources by companies and other organizations without proper authorization from countries and people concerned with compensation amount

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Arrange the following events in correct order pertaining to sewage treatment plant?

- (A) Bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide.
- (B) Primary effluent is passed into aeration tank for the growth of microbes into flocs
- (C) A small part of activated sludge is pumped back into aeration tank as inoculum and remaining to anaerobic sludge digesters.
- (D) The bacteria and fungi in the sludge are digested by another kind of bacteria

Choose the correct answer from the options given below:

1. (B), (C), (D), (A)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.10 Baculoviruses are pathogen that attack :

- 1. Reptiles
- 2. Plants
- 3. Insects
- 4. Birds

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.11 Lactational amenorrhea refers to:

- 1. Absence of menstruation
- 2. Absence of menstruation during intense lactation
- 3. No breast feeding
- 4. Intense breast feeding

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.12 How many different types of gametes are possible to be formed by an individual if its genotype for a particular trait is AABbccDd?

- 1. 4
- 2. 8
- 3. 16
- 4. 32

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.13 The assisted reproductive technique where embryos with more than 8 blastomeres are transferred into the human uterus is known as :

- 1. ICSI
- 2. ZIFT
- 3. GIFT
- 4. IUT

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.14 Bioactive molecule, cyclosporin A, is produced by :

- 1. *Monascus purpureus*
- 2. *Trichoderma polysporum*
- 3. *Clostridium butylicum*
- 4. *Acetobacter aceti*

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.15 Lichens are an example of -

- 1. Commensalism
- 2. Parasitism
- 3. Mutualism
- 4. Competition

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Those plants whose genes have been altered by manipulation are known as _____

1. Pest resistant plants
2. Genetically modified plants
3. Vectors
4. Plant Crystals

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Which of the following statements are correct with respect to T.H. Morgan?

- (A) His experiment led to discovering the basis for the variation that sexual reproduction produced.
(B) He mapped the position of genes on the chromosomes.
(C) He coined the term linkage.
(D) He crossed yellow-bodied, white-eyed *Drosophila* females to brown-bodied red-eyed *Drosophila* males and observed the results contrary to that of Mendel's.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Which one of the following hormones is related to spermatogenesis?

1. Androgens
2. Estrogen
3. Progesterone
4. LH

Options 1. 1

2. 2
3. 3
4. 4

Q.19 India has more than 1000 varieties of mango. This is an example of:

1. Species diversity
2. Genetic diversity
3. Family diversity
4. Ecological diversity

Options 1. 1

2. 2
3. 3
4. 4

Q.20 Which one of the following antibodies is released through colostrum?

1. IgM
2. IgA
3. IgE
4. IgG

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Arrange the following features related to decomposition cycle in a terrestrial ecosystem in correct sequence?

- (A) Fallen leaves eaten by insects and animals
- (B) Trees grow in the soil
- (C) Some nutrients leach into the soil
- (D) Decomposition by earthworm and bacteria etc.

Choose the correct answer from the options given below:

- 1. (B), (A), (D), (C)
- 2. (A), (B), (C), (D)
- 3. (A), (D), (B), (C)
- 4. (D), (C), (A), (B)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.22 Which one of the following hormones is responsible for ovulation?

- 1. Luteinizing hormone
- 2. Estrogen
- 3. Progesterone
- 4. Follicle stimulating hormone

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.23 Arrange the following cells in correct sequence of oogenesis.

- (A) Secondary oocyte
- (B) Oogonia
- (C) Primary oocyte
- (D) Ovum

Choose the correct answer from the options given below:

- 1. (B), (C), (A), (D)
- 2. (A), (B), (C), (D)
- 3. (B), (A), (D), (C)
- 4. (D), (C), (A), (B)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.24 The causes of biodiversity loss are:

- (A) Habitat loss
- (B) Invasion of alien species
- (C) Co-extinction
- (D) Endemism

Choose the *correct* answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.25 Which one of the following drinks are produced by distillation?

- (A) Whisky
- (B) Brandy
- (C) Wine
- (D) Beer

choose the correct answer from the option given below:

- 1. (A) and (B) only
- 2. (A) and (C) only
- 3. (B), (C) and (D) only
- 4. (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Arrange the following wall layers of microsporangium starting from outer to inner.

- (A) Tapetum
- (B) Endothecium
- (C) Middle layer
- (D) Epidermis

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (D), (B), (C), (A)
- 3. (B), (A), (D), (C)
- 4. (A), (D), (B), (C)

Options 1. 1

2. 2

3. 3

4. 4

Q.27 Which one of the followings represent a correct form of mature angiosperm embryo sac?

- 1. 8 celled with 7 nuclei
- 2. 7 celled with 8 nuclei
- 3. 7 celled with 7 nuclei
- 4. 8 celled with 8 nuclei

Options 1. 1

2. 2

3. 3

4. 4

Q.28 The blastocyst becomes embedded in the endometrium of the uterus. This is called :

- 1. Lactation
- 2. Parturition
- 3. Implantation
- 4. Gestation

Options 1. 1

2. 2

3. 3

4. 4

Q.29 *Entamoeba histolytica* is a protozoan parasite found in which part of human body?

- 1. Lungs
- 2. Liver
- 3. Small intestine
- 4. Large intestine

Options 1. 1

2. 2

3. 3

4. 4

Q.30 Match List-I with List-II

List-I

List-II

Term

Feature

(A)

(I) Flowers do not open

Autogamy

(B)

(II) Transfer of pollen grains from anther to stigma of different plant

Geitonogamy

y

List-I

(C)
Xenogamy

List-II

(III) Transfer of pollen grains from anther to stigma of another flower of same plant

(D)
Cleistogamy

(IV) Transfer of pollen grains from anther to stigma of same flower

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Match List-I with List-II

List-I

List-II

Term

Explanation

(A) Natality (I) Number of the same species individuals that have come into the habitat

(B) Immigration (II) Number of deaths during a given time in a population

(C) Mortality (III) Number of the same species individuals that have left the habitat

(D) Emigration (IV) Number of births during a given time in a population

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Arrange the following historical events in the history of life in chronological order?

(A) Natural selection

(B) Self-replicating molecules

(C) Eukaryotic cell

(D) Prokaryotic cell

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Match List-I with List-II

List-I

List-II

Term

Characteristics

(A) Bt toxin (I) ADA deficiency

(B) RNAi (II) Increased food supply

(C) Green revolution (III) cryI_{Ac} gene

(D) Gene Therapy (IV) mRNA silencing

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Chromosomal theory of inheritance was given by:

1. Mendel and de Vries
2. Watson and Crick
3. Sutton and Boveri
4. Beadle and Tatum

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Arrange the following spermatogenic cells in correct sequence of spermatogenesis.

- (A) Spermatids
- (B) Spermatogonium
- (C) Secondary Spermatocytes
- (D) Primary Spermatocytes

Choose the correct answer from the options given below:

1. (B), (D), (C), (A)
2. (D), (B), (A), (C)
3. (A), (B), (C), (D)
4. (C), (B), (A), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Arrange the following ducts pertaining to male reproductive system starting from seminiferous tubules

- (A) Vas deferens
- (B) Vasa efferentia
- (C) Rete testis
- (D) Epididymis

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (C), (D), (A)
3. (D), (A), (B), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.37 The first menstruation which begins at puberty is called :

1. Menopause
2. Menarche
3. Menstrual cycle
4. Ovulation

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Select the components that are needed to set up a polymerase chain reaction:

- (A) Two primers and dNTPs
- (B) Template DNA
- (C) DNA ligase
- (D) DNA polymerase

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.39 Chromosomes 'Z' and 'W' for sex determination are seen in :

- 1. Human
- 2. Grasshopper
- 3. Birds
- 4. *Drosophila*

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.40 Match List-I with List-II

List-I

List-II

Genetic disorder

Characteristics

(A) Haemophilia

(I) Mutation in HBB gene

(B) Thalassemia

(II) Sex-linked recessive disease

(C) Phenylketonuria

(III) Aneuploidy

(D) Klinefelter's syndrome

(IV) Mental retardation and reduced hair and skin pigmentation

Choose the correct answer from the options given below:

- 1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 2. (A) - (I), (B) - (III), (C) - (I), (D) - (IV)
- 3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.41 Read the following passage carefully.

Restriction enzyme

Restriction enzymes cut the strand of DNA a little away from the center of the **palindrome** sites, but between the same two bases on the opposite strands. This leaves single stranded portions at the ends. There are overhanging stretches called sticky ends on each strand. These are named so because they form hydrogen bonds with their complementary cut counterparts. This stickiness of the ends facilitates the action of the enzyme DNA ligase. The word palindromes refer to group of letters that forms the same words when read both forward and backward. Restriction endonucleases are used in genetic engineering as molecular scissors.

Overhanging stretches on each strand is called:

- 1. Hydrogen bond
- 2. Sticky end
- 3. Ligase
- 4. Palindrome

Options 1. 1

- 2. 2

3. 3

4. 4

Q.42 Read the following passage carefully.

Restriction enzyme

Restriction enzymes cut the strand of DNA a little away from the center of the palindrome sites, but between the same two bases on the opposite strands. This leaves single stranded portions at the ends. There are overhanging stretches called sticky ends on each strand. These are named so because they form hydrogen bonds with their complementary cut counterparts. This stickiness of the ends facilitates the action of the enzyme DNA ligase. The word palindromes refer to group of letters that forms the same words when read both forward and backward. Restriction endonucleases are used in genetic engineering as molecular scissors.

Which one of the following option is correct about restriction enzyme?

1. It helps in binding of DNA.
2. This is known as molecular scissors.
3. It has ability to multiply copies of plasmid.
4. It helps in transfer of piece of DNA.

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the following passage carefully.

Restriction enzyme

Restriction enzymes cut the strand of DNA a little away from the center of the palindrome sites, but between the same two bases on the opposite strands. This leaves single stranded portions at the ends. There are overhanging stretches called sticky ends on each strand. These are named so because they form hydrogen bonds with their complementary cut counterparts. This stickiness of the ends facilitates the action of the enzyme DNA ligase. The word palindromes refer to group of letters that forms the same words when read both forward and backward. Restriction endonucleases are used in genetic engineering as molecular scissors.

Which bond forms on sticky end of each strand?

1. Peptide bond
2. Hydrogen bond
3. Sulphur bond
4. Carbon bond

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the following passage carefully.

Restriction enzyme

Restriction enzymes cut the strand of DNA a little away from the center of the palindrome sites, but between the same two bases on the opposite strands. This leaves single stranded portions at the ends. There are overhanging stretches called sticky ends on each strand. These are named so because they form hydrogen bonds with their complementary cut counterparts. This stickiness of the ends facilitates the action of the enzyme DNA ligase. The word palindromes refer to group of letters that forms the same words when read both forward and backward. Restriction endonucleases are used in genetic engineering as molecular scissors.

Which one of the following enzyme helps to join DNA?

1. DNA polymerase
2. RNA polymerase
3. DNA ligase
4. Endonuclease

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Read the following passage carefully.

Restriction enzyme

Restriction enzymes cut the strand of DNA a little away from the center of the palindrome sites, but between the same two bases on the opposite strands. This leaves single stranded portions at the ends. There are overhanging stretches called sticky ends on each strand. These are named so because they form hydrogen bonds with their complementary cut counterparts. This stickiness of the ends facilitates the action of the enzyme DNA ligase. The word palindromes refer to group of letters that forms the same words when read both forward and backward. Restriction endonucleases are used in genetic engineering as molecular scissors.

Palindromes are:

1. The same group of letters when read both forward and backward
2. The different group of letters when read both forward and backward
3. Same group of letters when read only forward
4. Same group of letters when read only backward

Options 1. 1

2. 2

3. 3

4. 4

Q.46 MONOHYBRID CROSS

The inheritance of flower color in the dog flower (Snapdragon or *Antirrhinum sp.*) can be understood by crossbreeding between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr). When they were crossed, F_1 (Rr) was pink. When F_1 was self-pollinated, the F_2 results were Red, Pink and White. Here the genotype ratios were exactly as expected, but the phenotype ratios have been changed. What happened was that R was not completely dominant over r and this made it possible to distinguish Rr as pink from RR (Red) and rr (White).

In the above passage, the inheritance is an example of:

1. Co-dominance
2. Dominance
3. Incomplete dominance
4. Recessive trait

Options 1. 1

2. 2

3. 3

4. 4

Q.47 MONOHYBRID CROSS

The inheritance of flower color in the dog flower (Snapdragon or *Antirrhinum sp.*) can be understood by crossbreeding between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr). When they were crossed, F_1 (Rr) was pink. When F_1 was self-pollinated, the F_2 results were Red, Pink and White. Here the genotype ratios were exactly as expected, but the phenotype ratios have been changed. What happened was that R was not completely dominant over r and this made it possible to distinguish Rr as pink from RR (Red) and rr (White).

What will be the percentage of progeny in F_1 generation when RR (red flowered plant) is crossed with rr (white flowered plant) in the above passage?

1. 100% Rr
2. 75% RR and 25% rr
3. 25% RR, 50% Rr and 25% rr
4. 25% RR and 75% rr

Options 1. 1

2. 2

3. 3

4. 4

Q.48 MONOHYBRID CROSS

The inheritance of flower color in the dog flower (Snapdragon or *Antirrhinum sp.*) can be understood by crossbreeding between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr). When they were crossed, F_1 (Rr) was pink. When F_1 was self-pollinated, the F_2 results were Red, Pink and White. Here the genotype ratios were exactly

as expected, but the phenotype ratios have been changed. What happened was that R was not completely dominant over r and this made it possible to distinguish Rr as pink from RR (Red) and rr (White).

What will be the genotypic ratio in F_2 generation when F_1 (Rr) was self-pollinated?

1. 3:1
2. 2:1:1
3. 1:1
4. 1:2:1

Options 1. 1

2. 2
3. 3
4. 4

Q.49 MONOHYBRID CROSS

The inheritance of flower color in the dog flower (Snapdragon or *Antirrhinum sp.*) can be understood by crossbreeding between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr). When they were crossed, F_1 (Rr) was pink. When F_1 was self-pollinated, the F_2 results were Red, Pink and White. Here the genotype ratios were exactly as expected, but the phenotype ratios have been changed. What happened was that R was not completely dominant over r and this made it possible to distinguish Rr as pink from RR (Red) and rr (White).

What was the expected phenotypic ratio in any Mendelian monohybrid cross that had changed when F_1 (Rr) was self-pollinated in this cross?

1. 1:1
2. 1:2:1
3. 3:1
4. 1:3

Options 1. 1

2. 2
3. 3
4. 4

Q.50 MONOHYBRID CROSS

The inheritance of flower color in the dog flower (Snapdragon or *Antirrhinum sp.*) can be understood by crossbreeding between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr). When they were crossed, F_1 (Rr) was pink. When F_1 was self-pollinated, the F_2 results were Red, Pink and White. Here the genotype ratios were exactly as expected, but the phenotype ratios have been changed. What happened was that R was not completely dominant over r and this made it possible to distinguish Rr as pink from RR (Red) and rr (White).

What will be the phenotypic ratio in F_2 generation when F_1 (Rr) was self-pollinated?

1. 1:1
2. 1:2:1
3. 3:1
4. 1:3

Options 1. 1

2. 2
3. 3
4. 4